1 Battery input and protection

File: power_entry.kicad_sch

2 5V0 buck and 3V3 LDO

File: logic_power.kicad_sch

3 MCU, clocks, supplies, debug

File: mcu.kicad_sch

4 2.4 GHz RF path and antenna

File: rf.kicad_sch

5 Motor drive (DRV8711 + external FETs)

File: drive.kicad_sch

6 Motor connectors and encoder inputs

File: motors_encoders.kicad_sch

7 IMU and SPI flash

File: imu_flash.kicad_sch

8 User interface, expansion, test points

File: ui.kicad_sch

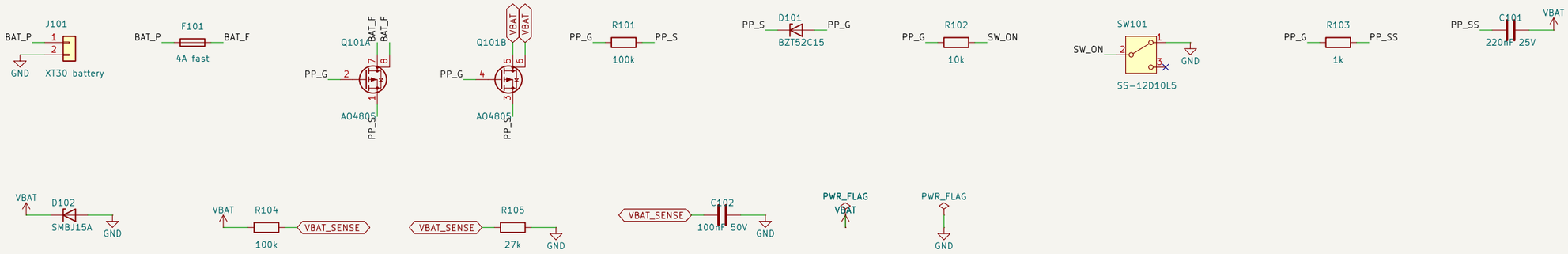
embedded-pro

Sheet: /

File: e-inverted-pendulum.kicad_sch

Title: e-inverted-pendulum chassis PCB

Size: A3	Date: 2026-11-04	**Rev: A**
KiCad E.D.A. 10.0.8		Id: 1/9



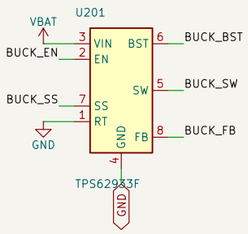
Generated by kicad/gen/build.py; do not hand-edit before the Phase 1 gate

embedded-pro

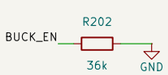
Sheet: /1 Battery input and protection/
File: power_entry.kicad_sch

Title: e-inverted-pendulum chassis PCB: Battery input and protection

Size: A3	Date: 2026-11-04	**Rev: A**
KiCad E.D.A. 10.0.8		Id: 2/9



















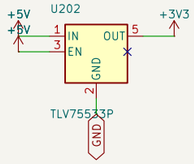


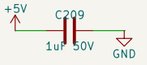
















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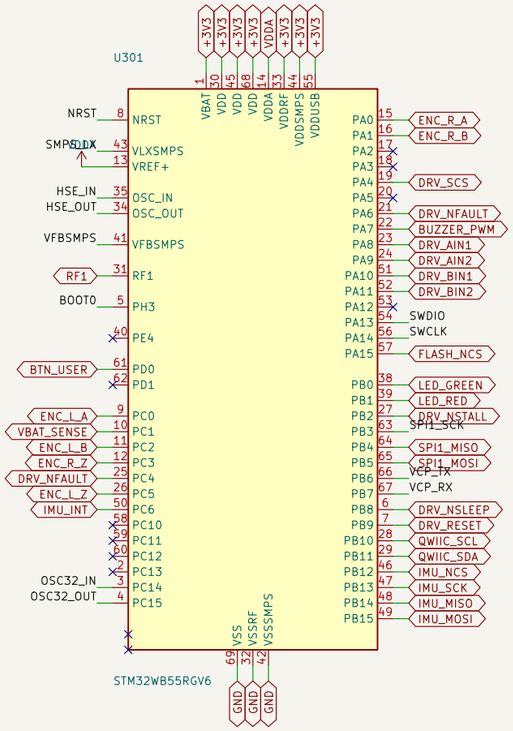
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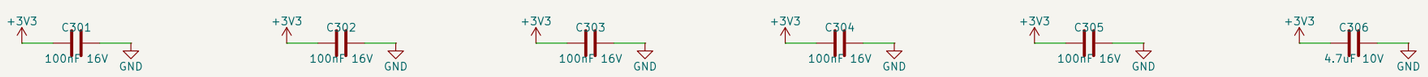
Sheet: /2 5V0 buck and 3V3 LDO/

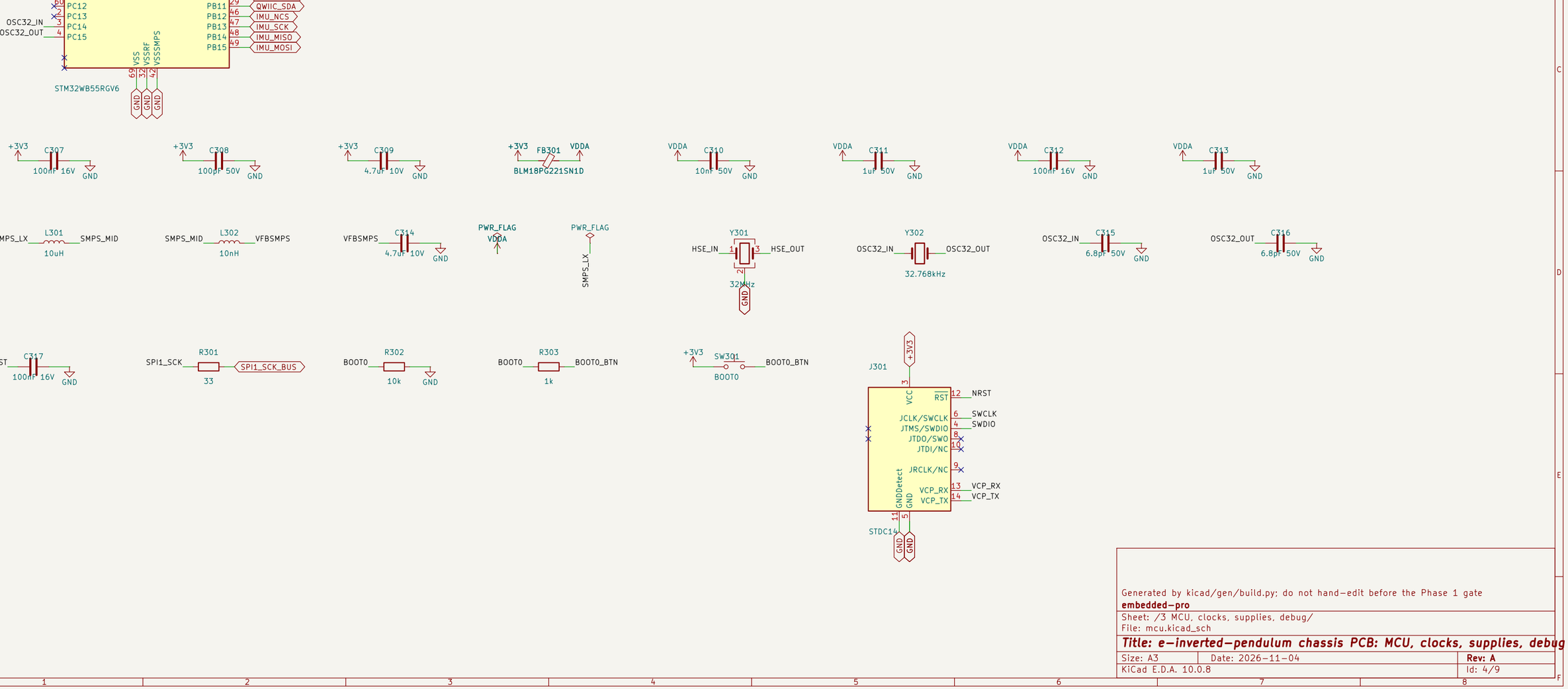
File: logic_power.kicad_sch

Title: e-inverted-pendulum chassis PCB: 5V0 buck and 3V3 LDO

Size: A3	Date: 2026-11-04	**Rev: A**
KiCad E.D.A. 10.0.8		Id: 3/9







Generated by kicad/gen/build.py; do not hand-edit before the Phase 1 gate

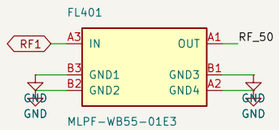
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Sheet: /3 MCU, clocks, supplies, debug/

File: mcu.kicad_sch

Title: e-inverted-pendulum chassis PCB: MCU, clocks, supplies, debug

Size: A3	Date: 2026-11-04	**Rev: A**
KiCad E.D.A. 10.0.8		Id: 4/9

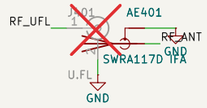










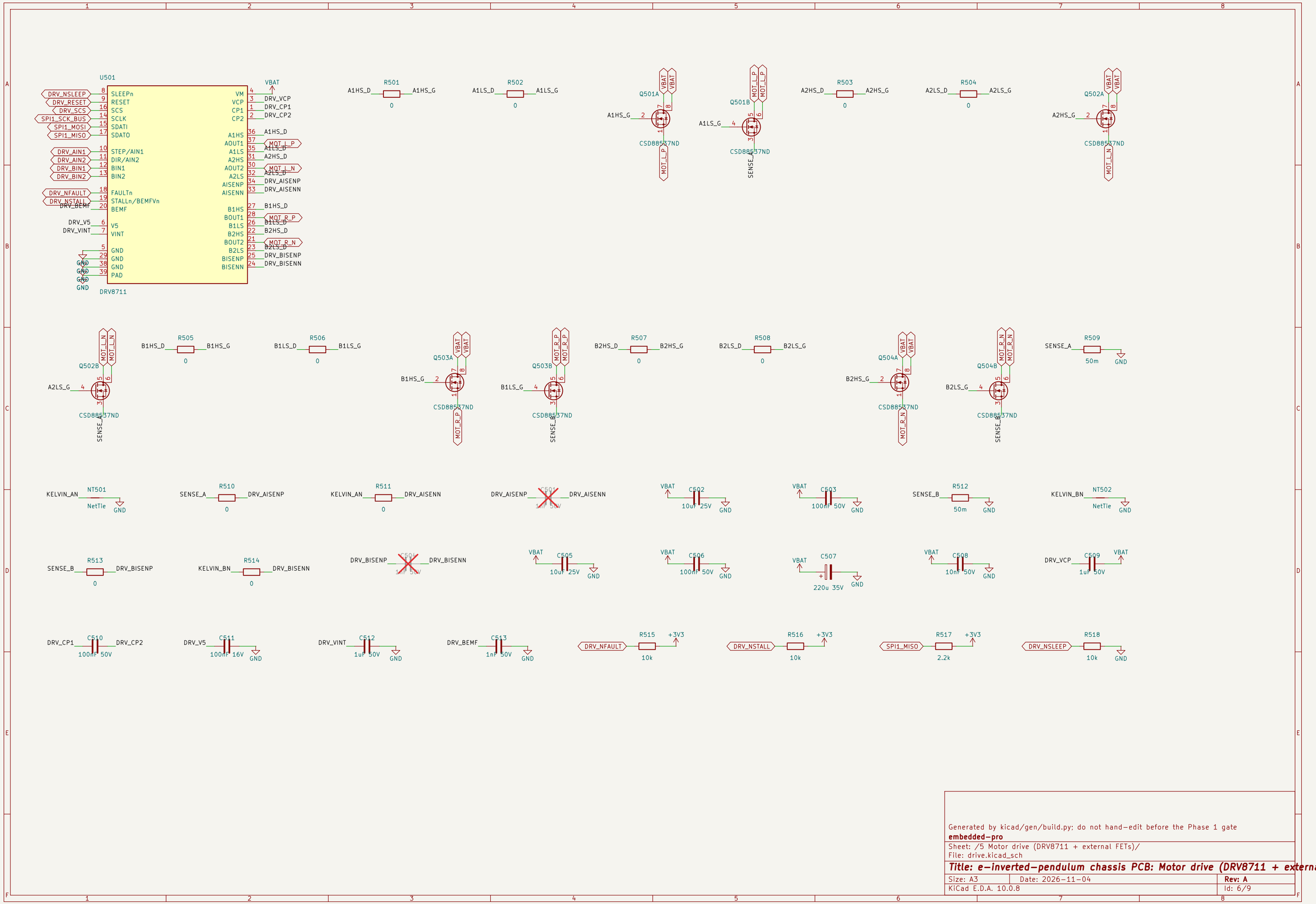


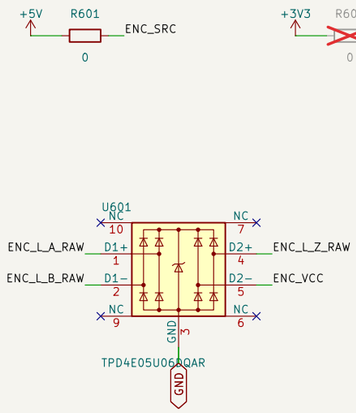
Generated by kicad/gen/build.py; do not hand-edit before the Phase 1 gate
embedded-pro

Sheet: /4 2.4 GHz RF path and antenna/
File: rf.kicad_sch

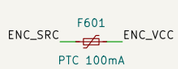
Title: e-inverted-pendulum chassis PCB: 2.4 GHz RF path and antenna

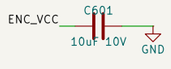
Size: A3	Date: 2026-11-04	**Rev: A**
KiCad E.D.A. 10.0.8		Id: 5/9

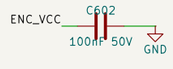

U501
DRV8711
SLEEPn
RESET
SCS
SCLK
SDATI
SDATO
STEP/AIN1
DIR/AIN2
BIN1
BIN2
FAULTn
STALLn/BEMFVn
BEMF
V5
VINT
GND
PAD
VM
VCP
CP1
CP2
A1HS
AOUT1
A1LS
A2HS
AOUT2
A2LS
AISENP
AISENN
B1HS
BOUT1
B1LS
B2HS
BOUT2
B2LS
BISENP
BISENN
DRV_NSLEEP
DRV_RESET
DRV_SCS
SPI1_SCK_BUS
SPI1_MOSI
SPI1_MISO
DRV_AIN1
DRV_AIN2
DRV_BIN1
DRV_BIN2
DRV_NFAULT
DRV_NSTALL
DRV_BEMF
DRV_V5
DRV_VINT
VBAT
DRV_VCP
DRV_CP1
DRV_CP2
A1HS_D
MOT_L_P
A1LS_D
A2HS_D
MOT_L_N
A2LS_D
DRV_AISENP
DRV_AISENN
B1HS_D
MOT_R_P
B1LS_D
B2HS_D
MOT_R_N
B2LS_D
DRV_BISENP
DRV_BISENN
R501
A1HS_D
A1HS_G
0
R502
A1LS_D
A1LS_G
0
Q501A
Q501B
CSD88537ND
SENSE_A
R503
A2HS_D
A2HS_G
0
R504
A2LS_D
A2LS_G
0
Q502A
Q502B
R505
B1HS_D
B1HS_G
0
R506
B1LS_D
B1LS_G
0
Q503A
Q503B
R507
B2HS_D
B2HS_G
0
R508
B2LS_D
B2LS_G
0
Q504A
Q504B
SENSE_B
R509
SENSE_A
50m
GND
NT501
KELVIN_AN
NetTie
R510
SENSE_A
DRV_AISENP
0
R511
KELVIN_AN
DRV_AISENN
0
C501
DRV_AISENP
DRV_AISENN
1nF 50V
C502
VBAT
10uF 25V
GND
C503
100nF 50V
R512
SENSE_B
50m
NT502
KELVIN_BN
R513
SENSE_B
DRV_BISENP
0
R514
KELVIN_BN
DRV_BISENN
0
C504
DRV_BISENP
DRV_BISENN
1nF 50V
C505
10uF 25V
C506
100nF 50V
C507
220u 35V
C508
10nF 50V
C509
DRV_VCP
1uF 50V
C510
DRV_CP1
DRV_CP2
100nF 50V
C511
DRV_V5
100nF 16V
C512
DRV_VINT
1uF 50V
C513
DRV_BEMF
1nF 50V
R515
DRV_NFAULT
+3V3
10k
R516
DRV_NSTALL
10k
R517
SPI1_MISO
2.2k
R518
DRV_NSLEEP
10k
Generated by kicad/gen/build.py; do not hand-edit before the Phase 1 gate
embedded-pro
Sheet: /5 Motor drive (DRV8711 + external FETs)/
File: drive.kicad_sch
Title: e-inverted-pendulum chassis PCB: Motor drive (DRV8711 + external F
Size: A3
Date: 2026-11-04
Rev: A
KiCad E.D.A. 10.0.8
Id: 6/9



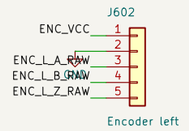






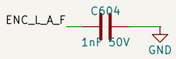




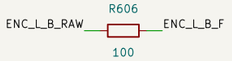


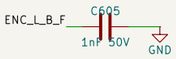






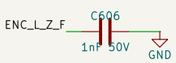


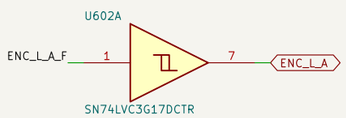


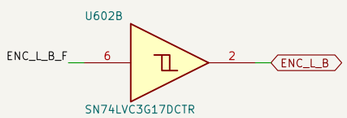


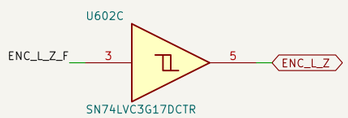


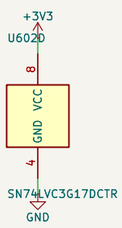


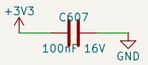


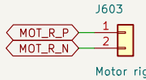




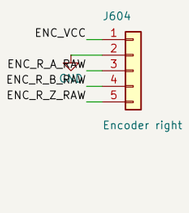


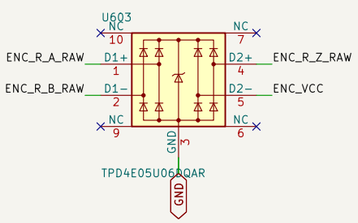


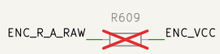




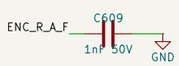






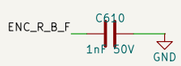






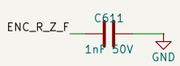


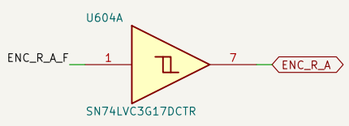


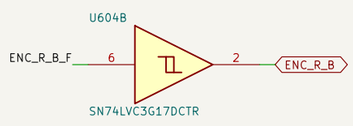


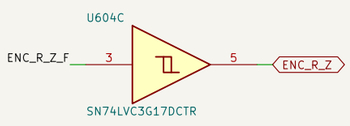


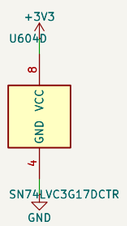














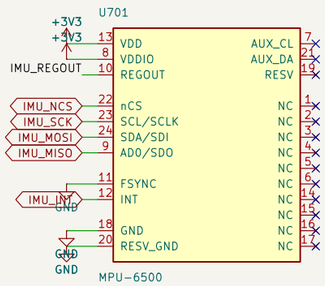
Generated by kicad/gen/build.py; do not hand-edit before the Phase 1 gate

embedded-pro

Sheet: /6 Motor connectors and encoder inputs/
File: motors_encoders.kicad_sch

Title: e-inverted-pendulum chassis PCB: Motor connectors and encoder inpu

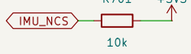
Size: A3	Date: 2026-11-04	**Rev: A**
KiCad E.D.A. 10.0.8		Id: 7/9

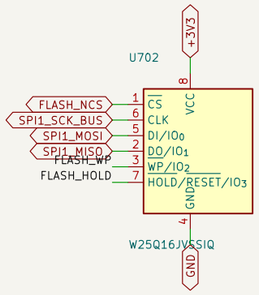




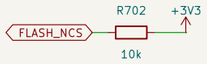
















Generated by kicad/gen/build.py; do not hand-edit before the Phase 1 gate

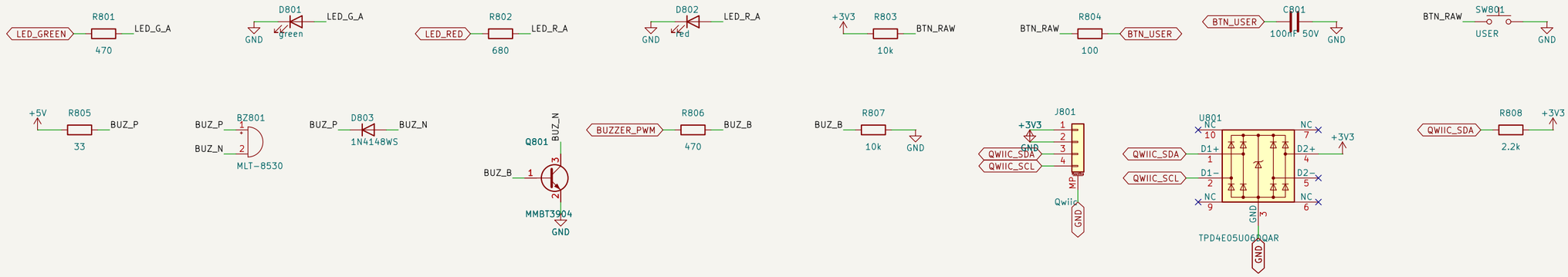
embedded-pro

Sheet: /7 IMU and SPI flash/

File: imu_flash.kicad_sch

Title: e-inverted-pendulum chassis PCB: IMU and SPI flash

Size: A3	Date: 2026-11-04	**Rev: A**
KiCad E.D.A. 10.0.8		Id: 8/9



Generated by kicad/gen/build.py; do not hand-edit before the Phase 1 gate

embedded-pro

Sheet: /8 User interface, expansion, test points/

File: ui.kicad_sch

Title: e-inverted-pendulum chassis PCB: User interface, expansion, test poin

Size: A3	Date: 2026-11-04	**Rev: A**
KiCad E.D.A. 10.0.8		Id: 9/9